

Department of Computer Science and Engineering

Course: CSE356 (Software Engineering Sessional) • Session: 2022–2023

LAB ASSIGNMENT REPORT

UML Design Specifications for MaximaPC

SUBMITTED BY (GROUP MEMBERS)

1. Md. Atikur Rahman
2. MD. Tazrin Hasan

ID: 22CSE017
ID: 22CSE038

Dept. of CSE, GSTU • Date: August 31, 2026

SUBMITTED TO

Dr. Mrinal Kanti Baowaly
Professor, Department of CSE
Gopalganj Science and Technology University

• 1. INTRODUCTION & ARCHITECTURAL CONTEXT

MaximaPC is a PC hardware and accessories e-commerce platform with an integrated custom PC Builder. The system is intended to let customers browse and search products, assemble compatible computer configurations, compare products, manage a wishlist, purchase components through an integrated payment gateway, save builds, review products, and track orders. Administrators manage the product catalogue, stock, users, and orders.

Following software engineering design principles, software design bridges the gap between requirement specifications and source implementation. This report formally defines three complementary UML perspectives:

- **Use Case Diagram:** Captures Guest, Registered User, and Admin capabilities, the system boundary, and interactions with Google OAuth and the payment gateway.
- **Sequence Diagrams (5 Scenarios):** Models chronological message exchanges and alternate execution flows for authentication, compatibility validation, checkout/payment, CSV import, and build persistence.
- **Class Diagram:** Establishes the principal e-commerce domain classes, their attributes, operations, and multiplicity associations.

• 2. USE CASE DIAGRAM

The Use Case Diagram models interactions between the three principal actors (**Guest**, **Registered User**, **Admin**) and the MaximaPC system boundary. Guests can browse the catalogue, view products, use the PC Builder, and proceed through cart and checkout; registered users gain persistent features such as comparison, wishlist management, saved builds, order history, and reviews. The diagram also identifies **Google OAuth** and the **Payment Gateway** as external participants in authentication and payment respectively.

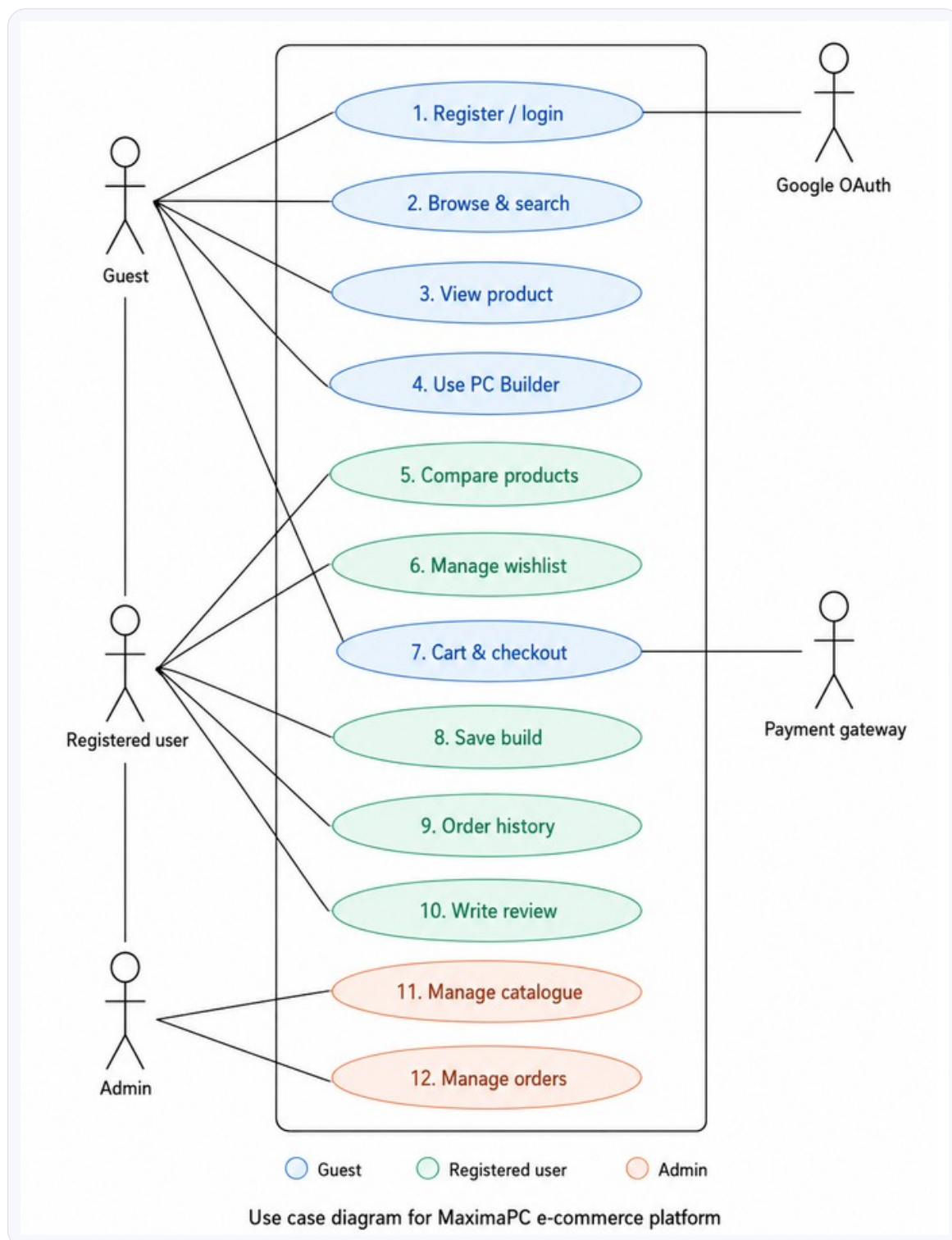


Figure 1: Use Case Diagram of the MaximaPC E-Commerce Platform

3.1 SEQUENCE DIAGRAM 1: USER AUTHENTICATION – LOCAL LOGIN & GOOGLE OAUTH

Models the two authentication paths supported by MaximaPC. In the local path, the server retrieves the stored password hash and verifies the supplied password using `bcrypt.compare()`. In the Google OAuth path, the browser is redirected to Google, an authorization callback is received, the profile is returned, and the server looks up the account using Google identity/email before creating an account when necessary. Both paths finish by establishing a session cookie.

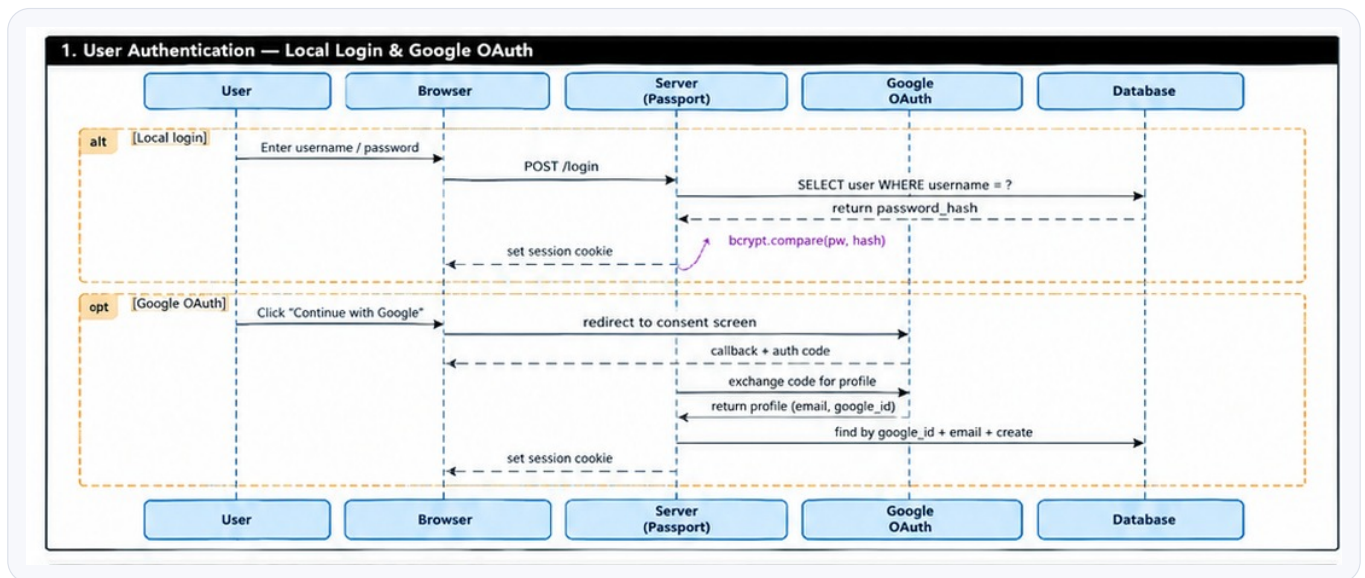


Figure 2: Sequence Diagram 1 – User Authentication with Local Login and Google OAuth

3.2 SEQUENCE DIAGRAM 2: PC BUILDER – REAL-TIME COMPATIBILITY VALIDATION

Illustrates the distinctive MaximaPC Builder workflow. When a user selects a component, the Builder UI obtains its specification attributes and passes the selected components to the client-side compatibility engine. The engine checks relevant properties such as socket, RAM type, and form factor. The alternate flow either updates the slot when compatible or flags the selected slot when incompatible.

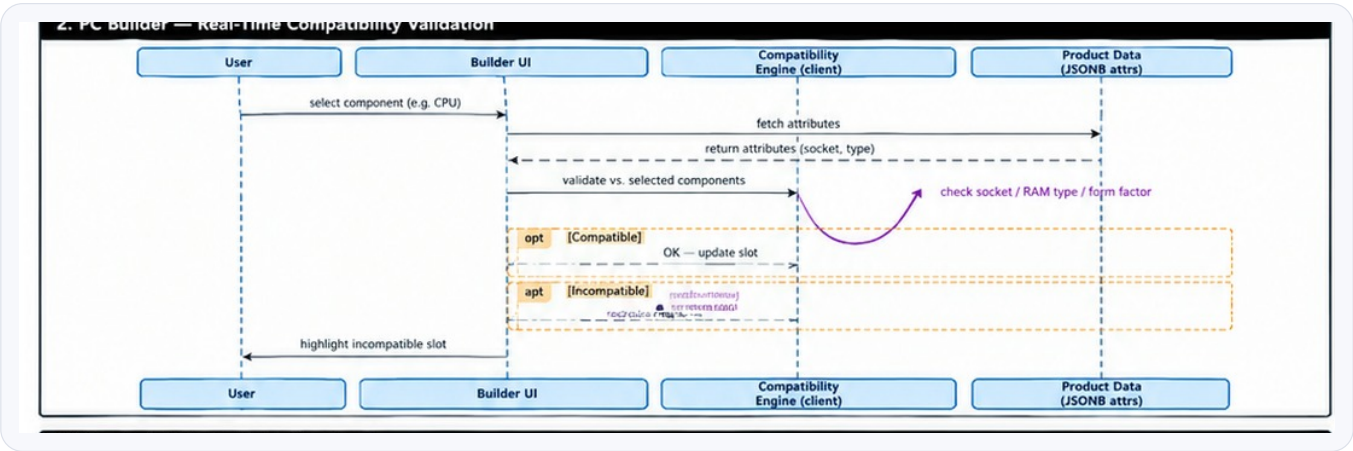


Figure 3: Sequence Diagram 2 – PC Builder Real-Time Compatibility Validation

3.3 SEQUENCE DIAGRAM 3: CHECKOUT & PAYMENT (SSLCOMMERZ)

Models the multi-system checkout workflow. The user submits the checkout form, the application creates an order with a pending status, and a payment session is initiated with SSLCommerz. The user is redirected to the payment gateway and submits payment details there. The gateway callback is processed by the application, which updates the order status to *paid* or *failed* and returns the user to the confirmation flow.

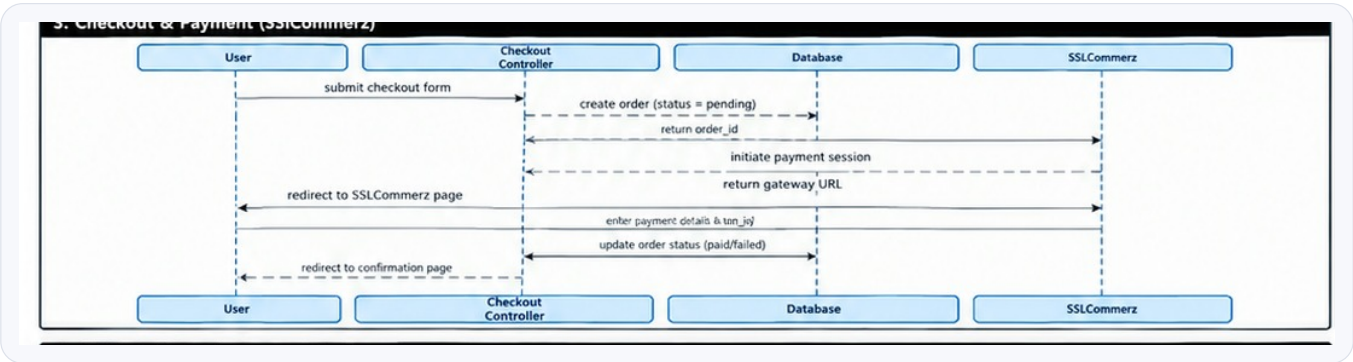


Figure 4: Sequence Diagram 3 – Checkout and Payment through SSLCommerz

3.4 SEQUENCE DIAGRAM 4: ADMIN BULK PRODUCT IMPORT (CSV)

Shows the non-trivial administrative import workflow. An administrator uploads a CSV file through the upload handler (multer), the CSV parser processes the file, and each row is validated before insertion. Valid rows are inserted as products while invalid rows are collected with an error reason. The final import summary reports how many records were imported and how many failed.

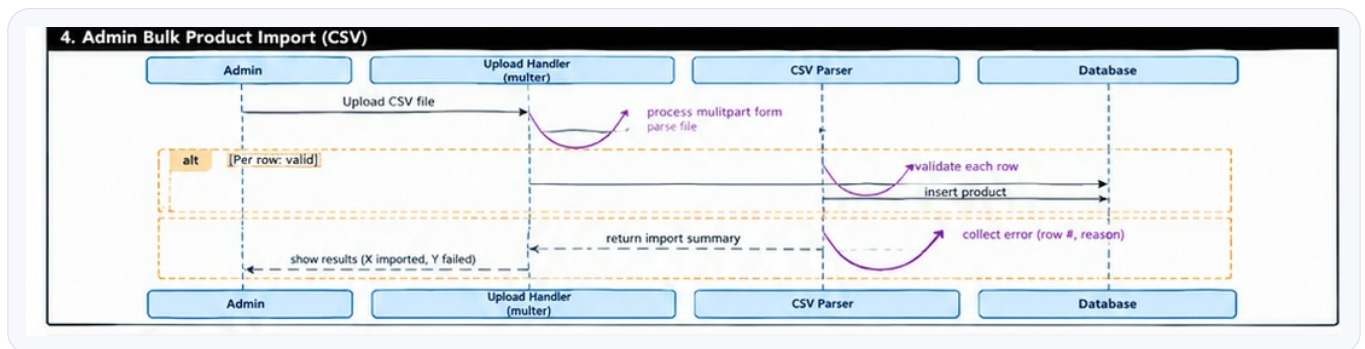


Figure 5: Sequence Diagram 4 – Admin Bulk Product Import

3.5 SEQUENCE DIAGRAM 5: SAVE / RELOAD A BUILD

Models persistence of a custom PC configuration. On save, the Builder UI sends the build name and component list to the server; a build record is inserted first, followed by multiple build_items rows associated with the generated build_id. During reload, the server retrieves the saved build and joins the build items with product information, returning the stored attributes so that the Builder UI can reconstruct the configuration slot by slot.

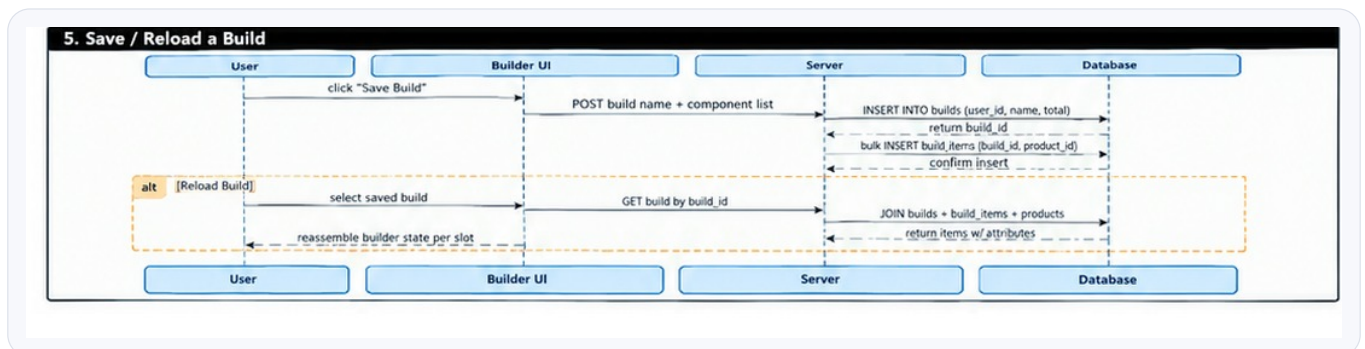


Figure 6: Sequence Diagram 5 – Save and Reload a PC Build

• 4. CLASS DIAGRAM (STATIC DOMAIN STRUCTURE)

The Class Diagram represents the principal domain structure of MaximaPC and is aligned with the relational entities used by the e-commerce platform. It covers **User**, **Build**, **BuildItem**, **Order**, **OrderItem**, **Product**, **Category**, **Wishlist**, and **Review**. The model shows attributes, core operations, foreign-key relationships, and multiplicity between users, builds, products, orders, and related records. Wishlist and Review are marked as planned in the supplied design because they belong to the target feature scope even where they are not part of the original seven-table baseline.

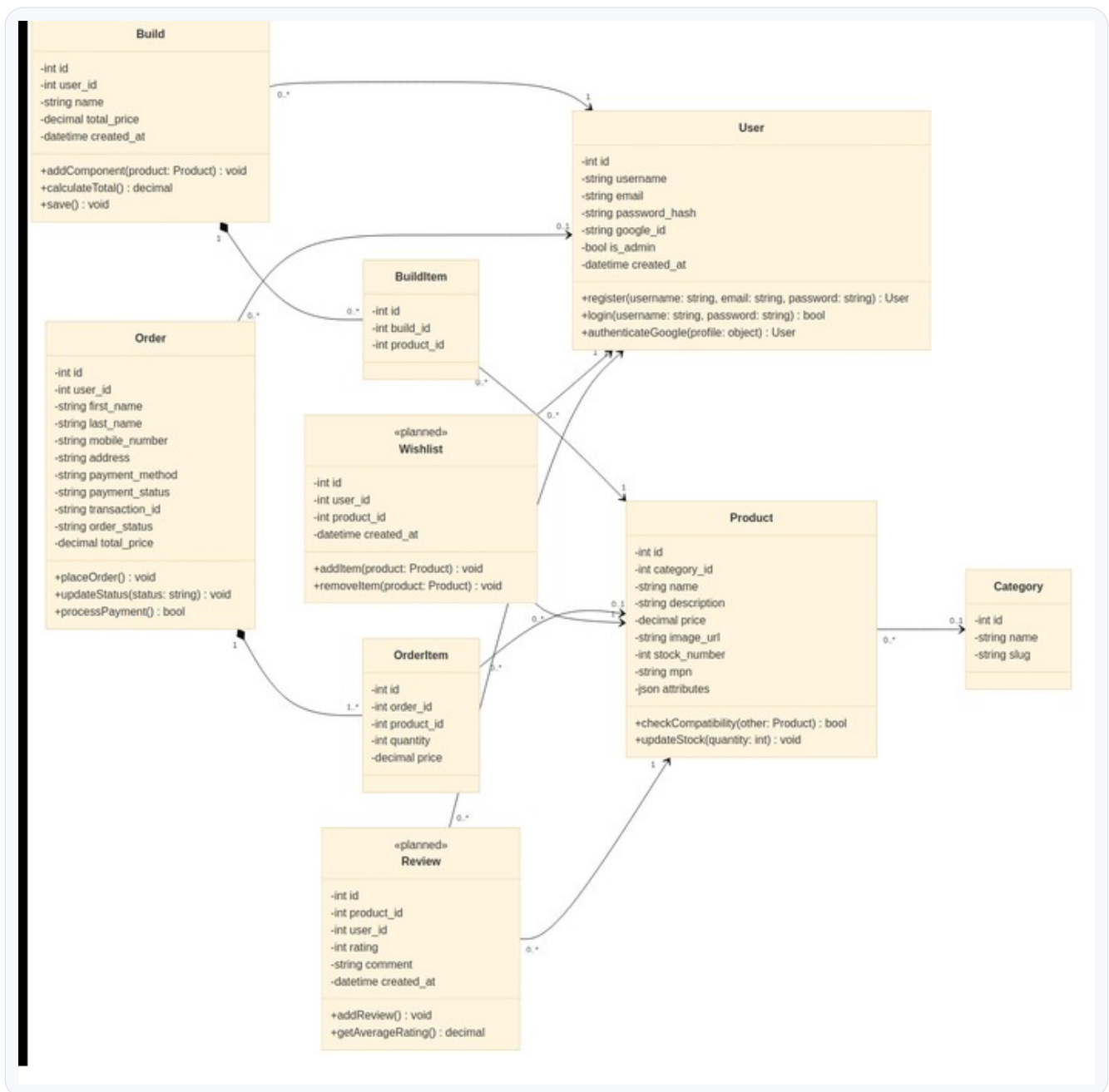


Figure 9: Class Diagram of the MaximaPC E-Commerce Platform

• 5. SOFTWARE DESIGN PRINCIPLES EVALUATION

The MaximaPC UML models are evaluated against software design principles taught in CSE356:

Principle	Core Objective	Implementation in MaximaPC
High Cohesion	Single Responsibility Principle (SRP) per component.	The compatibility engine focuses on component validation; checkout coordinates order/payment processing; and the CSV parser focuses on structured product import and row-level validation.
Low Coupling	Modules communicate through minimal, necessary data.	Controllers and UI flows exchange focused values such as user IDs, product IDs, build IDs, and order IDs instead of exposing complete database records. External payment communication is isolated to the checkout/payment flow.
Information Hiding	Encapsulating private implementation details.	Password hashes, JSONB product attributes, persistence details, and payment-gateway communication are kept behind the relevant authentication, product, database, and checkout operations.

• 6. CONCLUSION

Together, the Use Case Diagram, five comprehensive Sequence Diagrams, and Class Diagram provide complementary static and dynamic architectural models for the MaximaPC system. The models cover the principal actors and capabilities, the most interaction-intensive workflows, and the core relationships among users, products, builds, orders, and supporting e-commerce entities. They therefore provide a structured UML design view suitable for the CSE356 Software Engineering Sessional submission.